# Investigating the Effects of Playing a Targeted 2D Directional-Logic Game on the Improvement of Foundational Algorithmic Thinking Skills in Early Primary School Children Aged 7 to 9

# An Experimental Study.

by

**Group 2**

Cagla Eraslan

Hilal Semiz

Yusuf Türkay

Bogazici University

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| Summary Table | |
|---|---|
| **Group number** | 2 |
| **Group members' names** | Çağla Eraslan, Hilal Semiz, Yusuf Türkay |
| **Your research topic statement** | Investigating the effects of playing a targeted 2D directional-logic game on the improvement of foundational algorithmic thinking skills in early primary school children aged 7–9. |
| **Your directional research hypothesis** | Children aged 7–9 who participate in a 4-week targeted 2D directional logic game intervention will demonstrate significantly greater gains in foundational algorithmic thinking skills than children who receive traditional, non-game-based instruction. |
| **What is your independent variable?** | A targeted 2D directional-logic game intervention |
| **What is your dependent variable?** | Improvement in foundational algorithmic thinking skills. |
| **Clearly state your population** | Children aged 7–9 enrolled in public or private primary schools (early primary) |
| **Who are your participants?** | Children aged 7–9 from public/private educational institutions who meet inclusion criteria and have not previously played similar directional-logic games. |
| **What is your sample size?** | N = 36 total<br>Experimental group: n = 18 (6 High Ability, 6 Average Ability, 6 Low Ability)<br>Control group: n = 18 (6 High Ability, 6 Average Ability, 6 Low Ability) |
| **Sampling strategies** | Stratified random assignment at the participant level: participants stratified by ability level (high/average/low) and gender, then randomly assigned within each stratum to experimental vs. control groups (3 per condition per stratum). |
| **Design of your study** | Nonequivalent control group quasi-experimental pretest–posttest design<br>O X1 O<br>O X2 O |
| **Name of your data collection instrument(s)** | TechCheck2- Turkish Adaptation |
| **Reliability and validity data of your instrument(s)** | The Turkish adaptation of Tech Check 2 demonstrates acceptable internal consistency (KR-20 = .70; $\alpha \approx .70$). Item difficulty values range from .59 to .92, and item discrimination values range from .16 to .69, indicating appropriate difficulty and adequate discrimination for early primary students |
| **Materials used, if any** | The Sokoban puzzle game (Thinking Rabbit Inc., 1982) (accessed online via Math Is Fun: https://www.mathsisfun.com/games/sokoban.html ) |

# Declaration of Equal Contribution and Honesty

EŞİT KATKI VE DÜRÜSTLÜK BEYANI

*Aşağıdaki tablolardan en az birinde tüm imzaların tamamlanmış olması gereklidir. Aksi takdirde tüm grup üyeleri ayrı bir sınava alınacaktır.*

Bu ödev akademik dürüstlük çerçevesinde grup üyeleri tarafından hazırlanmıştır ve intihal içermez. Yapay zeka araçları izin verilen kısımlarda ve şekillerde kullanılmış ve kullanımı ile ilgili gereken bilgiler istenen şekilde sağlanmıştır.

Aşağıdaki imzalar bu beyanı güvence altına almaktadır.

Bu ödev tüm grup üyelerinin eşit katılımı ve katkısı ile tamamlanmıştır.

Tarih: 20.12.2025

| Grup Üyesi Adı Soyadı | İmza |
|---|---|
| Çağla Eraslan | |
| Hilal Semiz | |
| Yusuf Türkay | |

*Grup üyeleri ödeve eşit katkı yapmadılar ise aşağıda belirtilen tabloda uygun oran belirtilmelidir.*

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# The Literature Review

## Introduction

Although digital gaming has become a significant part of children's daily lives, there has not been sufficient empirical educational research examining how the various types of digital games may impact the development of algorithmic thinking and basic coding skills among young children. While coding continues to be addressed as a necessary 21st-century skill, much of the existing research on coding has been based on limited definitions of coding-related competencies, shorter time periods for experimentation, and varying data collection methods, resulting in very little empirical proof supporting the development of early algorithmic cognitive thinking skills. Several types of digital games that focus on using 2D directional logic have been found to be effective tools for enhancing children's Sequencing, Planning, and Predictions and Error Correction Skills. Still, there have not been any studies isolating these types of games for the purposes of developing Algorithmic Thinking in young children. Therefore, how often and how much of these types of games should be played by children to enhance their ability to develop their Algorithmic Thinking skills is still unknown.

The purpose of this study is to investigate the effects of playing targeted 2D directional-logic games on the improvement of foundational algorithmic thinking skills in early primary school children aged 7 to 9. Using a quasi-experimental, pretest–posttest nonequivalent control group design, the study investigates whether children who engage in structured Sokoban gameplay demonstrate greater improvements in algorithmic thinking skills than children who receive traditional, non-game-based instruction. This includes sequencing, path planning, multi-step strategy generation, and debugging. The independent variable in this research is the type of instructional experience (2D Directional-Logic Game-Based Intervention and Traditional Instruction) and the dependent variable will be the improvement of foundational algorithmic thinking skills, as assessed by using a standardized assessment.

## Evidence From The Literature

Computational thinking (CT) describes the mental capabilities that enable a student to identify problems, create an algorithm for solving this issue and measure the effectiveness of their algorithm. (Giannakoulas & Xinogalos, 2023, 2024). Some critical components of the CT framework are abstraction, sequencing, decomposition, debugging and algorithmic thinking. The AT component refers to developing a set of sequential instructions, creating

loops/conditional statements, and breaking tasks into smaller pieces for the purpose of creating functional solutions. Early researchers in the field of CT (Kanaki & Kalogiannakis, 2022a) highlighted the AT skill set. Also, studies in the field of game/digital learning have looked at the way these skills influence performance outcomes in educational video games. Research consistently indicates that young students develop strong foundational skills in computational thought processes through their engagement with educational video games during their first few years of school. (Dlab et al., 2020; Piedade & Dorotea, 2023).

Research has found that educational games can develop algorithm precision and abstraction in problem-solving, sequence construction, and loop building amongst students aged 6-10 years. (Giannakoulas & Xinogalos, 2023, 2024). Research indicates that games can assist children in creating stepwise sequences through loops, as well as by establishing conditional rules. (Giannakoulas & Xinogalos, 2023). In addition, the data from classroom research has demonstrated that digital games may support students in participating in active and involved collaboration when participating together by allowing students to share their rationale and work collaboratively to resolve problems. (Giannakoulas & Xinogalos, 2023). While the area of debugging and evaluation has grown in computer science (CT) research concerning game-based learning, it remains minimally investigated.

Motivation is essential for students to be motivated and participate in constructing knowledge (i.e. developing CT skills). Both groups of students that used game-based lessons developed higher levels of motivation and persistence than students who participated in traditional lessons (Hsu, 2018; Yang, 2024). Game-based activities that support learning through student-created questions and peer problem-solving improve the development of metacognitive awareness; therefore, students who participate in these types of activities will be able to correct misconceptions they may have (Hsu, 2018). Learning using game-oriented (GO) instruction frequently produces greater gains than traditional instruction, with enhanced problem-solving, sequencing, and logical reasoning capabilities gained through GO learning while considerably increasing confidence levels and participation from students (Yang, 2024). Two-dimensional puzzle games and directional-logic games have demonstrated success amongst young learners as game formats have motivated students to test their ideas, receive feedback, and understand the result of their selections. (Kanaki & Kalogiannakis, 2022b; Zengin & Karal, 2024).

Teacher scaffolding is also important for developing computational thinking (CT). Falloon and colleagues found that using planned and structured CT lessons, especially those using programmable robots, can help young children learn about sequencing, patterns, and

early debugging. When teachers guide students during gameplay, students develop a better understanding of CT than through unguided exploration. Wong, Zhao, & Xu found that by using block-style and unplugged games to introduce and demonstrate algorithmic concepts visually and simply, the cognitive demands placed on young children are less than when they are learning about algorithmic concepts without these types of games based on visual cues, and through the use of simple, visual representations. The use of teacher reflection, and feedback, during and after gameplay allows students to transfer their CT skills learned in one setting to different contexts. This concept fits within the GLAT framework, which describes how students progress from working with concrete activities to utilizing abstract cognitive reasoning (Dlab et al., 2020).

In addition to the GLAT framework expanding our knowledge of CT through the use of video games, there is also an increasing number of studies that provide evidence to support Digital Game Based Learning (DGBL) as well. Multiple studies indicate that short-term gaming interventions significantly improve children's abilities to utilize Algorithmic Thinking and Problem Solving techniques.(Falloon, 2024). Similar improvements can be noted among children ages 7 to 9, with these advances appearing to be associated with access to immediate feedback while engaging with a structured task in a game (Zengin & Karal, 2024). As reviewed in large studies, both serious and digital games provide opportunities for children to develop CT, programming skills, and motivation, with 2D gaming environments being particularly successful for young children (Gunderson & Lampropoulos, 2025). Game-based learning strategies outperform traditional instruction when accompanied by educational objectives that can be measured (Piedade & Dorotea, 2023).

Age, as we continue to develop mentally and physically, is another factor that impacts how well a person learns CT. In developing algorithmic thinking, it is seen that with an increase in age, this ability improves; as reflected in research findings indicating that cognitive maturity affects the ability of individuals to perform this skill (Kanaki & Kalogiannakis, 2022a). Likewise, research findings demonstrate that when children of equal learning resources were given identical types of opportunities, there were no observable differences in performance between genders. (Kanaki & Kalogiannakis, 2022b). In addition, research indicates that 2D logic-based digital games are valid for appropriate age and generation-based CT assessment, which is consistent with a number of other studies that demonstrate improvements in directional following, pattern recognition, sequencing and debugging abilities when CT activities for young learners are structured (Wong et al., 2024).

Theoretical studies conducted recently indicate that both the design of the game and the teaching method utilized are critical determinants of how effective game-based CT instruction can be for learners. To maximize learning, there must be clear learning goals, the provision of scaffolds, students must be provided with opportunities for reflection, and assessment measures that allow for the administration and application of CT skills outside of the game (Kovtaniuk et al., 2025). Additionally, the development of CT may be enhanced through systematic progression in the complexity of the tasks and activities presented to students, as illustrated in the GLAT Model (Dlab et al., 2020). Systematic literature reviews have shown that combining a gamified approach with high-quality pedagogy enhances the development of CT knowledge and learning attitude (Piedade & Dorotea, 2023).

The body of research indicates that there is a solid correlation between digital (online) learning and the improvement of computational thinking (CT) skills among youth when education has been designed to provide structure, include feedback, foster self-reflection and offer developmentally appropriate formats; however, there continues to be a need for further exploration into several areas, including debugging, evaluating and understanding the effect of genre-specific games on CT skills, and developing reliable tools for measuring CT skills. Therefore, there is an urgent need for additional studies on the effectiveness of structured, game-based CT instruction within the early primary classroom setting.

**Summary and Justification**

More and more research shows that digital and game-based learning environments can help to create computational thinking (CT) skills in young learners. CT has been studied for many years, though most studies look at CT as a broad category; very few researchers have isolated out the specific algorithmic skills that can be measured directly (Giannakoulas & Xinogalos, 2023, 2024). Additionally, empirical findings from the different studies in this area are not easily compared to one another due to the small sample sizes used in these studies, the short durations of the interventions, and the different tools used to assess CT (Giannakoulas & Xinogalos, 2024).

A limited amount of research has been published about the development of 2D Directional Logic Games even though some articles point to 2D Directional Logic Games as a suitable and effective means of developing CT for younger learners. (Kanaki & Kalogiannakis, 2022b). The majority of previous studies have addressed the educational aspects of the design, overlooking the potential contribution to CT skill development from entertainment-oriented, fun-based games, which are the types of games that children

regularly play in their home environment. Additionally, much of the existing literature on CT skill development was conducted with samples of children grouped across broad age categories (i.e., 6-10 years), masking the early age-specific changes that occur in foundational algorithmic skills as they are developed and continuously modified during early years. A focus on children aged 7 to 9 years will provide specific insight into how foundational algorithmic skills are developed and changed as children gain experience and knowledge in primary school.

Additionally, while game-based learning methodologies are frequently contrasted to other digital or 'gamified' implementations within the literature; few studies have compared the use of a targeted game-based experience against a standard (non-game), instructional methodology with regard to developing fundamental algorithmic thinking skills. As such, it is not possible to ascertain whether any gains made in algorithmic thinking can be distinctly linked to the specific affordances associated with game-based interactions versus simply being exposed to instructional content or benefitting from practice.

This research addresses the gaps identified in prior research on how children's algorithmic thought process develops via exposure to a specific 2-D directional logic-type of game. The goal of this research study is to determine how much benefit directional logic gameplay provides to students who complete a targeted intervention using directional logic (defined as the rules governing how to program) in comparison to those who receive the same length of time but participate only in classroom instruction. The study uses a quasi-experimental design, assessing student performance via pretest and post-test scoring using TechCheck2(Turkce) as a standardized, performance-based measure of foundational computational and algorithmic thinking skills

The inclusion of a traditional instructional control group allows for the most appropriate comparison of the effectiveness of game-based versus traditional approaches to developing computational thinking skills in young primary students.

In this respect, the findings from this study will not only be useful within the educational environment, but they will also provide a clearer view for parents on the attitudes that they develop through their children's experiences with fun or entertaining games and help provide a more informed perspective on how those types of games contribute to children's cognitive development.

## A Directional Hypothesis

Children aged 7–9 who participate in the 4-week 2D directional-logic game intervention will demonstrate significantly greater gains in foundational algorithmic thinking, as measured by Tech Check 2(TR) pretest–posttest scores, than children who receive traditional, non-game-based instruction.

# Method

## Study Design and Variables

### *Study Design*

The study employs a nonequivalent control group quasi-experimental pretest–posttest design. Both groups will complete pretests and posttests; the experimental group will participate in a 2D directional-logic game intervention, while the control group will receive traditional, non-game-based instruction

To reduce selection threats commonly associated with this design, participants were stratified by ability level and gender to balance baseline group composition and then randomly assigned within each stratum to experimental and control conditions.

(O = test,pre/post tests; X1= unusual/game treatment; X2= control treatment/ traditional lesson)

Represented as: O X1 O
O X2 O

### *Variables*

This research will test the following hypothesis related to the independent variable, 'group type'-which will consist of an experimental group playing a 2D logical game and a control group receiving traditional instruction-on the dependent variable 'improvement' as measured by pre-test and post-test scores.

## Participants & Sampling Procedures

The selection of participants will be guided by predetermined criteria for inclusion and exclusion based upon the age range of 7 to 9 years old who are enrolled in either public or private educational institutions. Children who have previously played games similar to those used for intervention will be excluded from the sample in order to remedy potential

effects due to prior exposure. In order to minimize any possible influences related to the child's gender or ability level, children's ability levels will be categorized in three groups (high, average, and low) prior to grouping them into two groups based on gender. Participants' ability levels will be categorized based on school achievement information (e.g., the most recent term grade) together with a teacher-rated three-level classroom performance rubric. Therefore, each of the above factors results in a total of six strata.

Once the six strata are defined to include a total of three randomly selected participants from each stratum into an experimental group and three participants from each stratum into a control group, a sample of 36 will be generated: 18 experimental group participants and 18 control group participants, with equal representation of the specified characteristics. The purpose of the stratified grouping is to increase the comparability of the groups and decrease the impact of selection bias, thereby increasing the internal validity of the research.

**Data Collection Instrument(s), Reliability, and Validity**

For this research the important instrument applied for the acquisition of data is Tech Check 2 (Turkish version). This is an already established and performance-based test designed for the assessment of the computational thinking capabilities of early primary school students. Actually, Tech Check 2 was originally designed and developed by Relkin et al. This test was further adapted into Turkish by authors Çetin, Şendurur, and Otu in the year 2022 specifically for the Turkish primary school students. This test evaluates significant aspects of computational thinking that include sequences, logical structuring, control structures (such as Repetition and Cause-and-Effect), debugging, and problem decomposition. This represents the dependent variable of the study.

There are a total of fifteen questions in Tech Check 2. The questions of Tech Check 2, like those of Tech Check 1, were created to measure basic algorithmic thinking skills using a dichotomous scoring system (i.e., correct = 1; incorrect = 0) with a maximum score of 15 (the higher the score, the better the individual's ability to think about algorithms). None of the tasks involve programming knowledge or use of any type of technology since they can all be completed without any technology or programming knowledge (unplugged activities). The reason for this is to enable the determination of whether or not an individual is capable of developing the ability to use Computational Thinking skills without any programming knowledge; therefore, all of the fifteen tasks were created with 7–9-year-old children in mind as potential test-takers.

In terms of the presentation and structure of items, Tech Check 2 consists of image and visual-based multiple-choice questions where the test-taker has four answer choices for each item and is tasked with using visual clues that illustrate or tell a story to solve the problem regarding the visual representation based on one logical rule and implementing the rule to take the appropriate actions to solve the problem. For example, in order to answer a question in Tech Check 2, the test-taker may have to select the correct sequence of actions or be able to identify an object in a visual array that could not be encoded or extracted. These types of questions primarily remove the need to read for the test-taker since the testing performance will measure an individual's ability to employ logic and to use algorithms rather than their reading ability.

The psychometric properties of Tech Check 2 were researched in the study of the Turkish adaptation. Internal consistency reliability was measured through the Kuder-Richardson Formula 20 (KR-20) because it was a formula apt for use in a dichotomous scale. From the data given, the KR-20 reliability coefficient of Tech Check 2 was given as 0.70 for third-grade students, making it reasonably reliable for research (Çetin et al., 2022). However, as the KR-20 formula is an equivalent formula to Cronbach's Alpha formula in dichotomous data, it means its reliability estimate can also be $\alpha = 0.70$.

Through the study of item level data, the test's reliability and validity can be confirmed. For third grade students, the difficulty values for each question on the test ranged from 0.59 (easiest) - 0.92 (hardest) with an average overall difficulty value of 0.72. This indicates that there were many questions of varying levels of difficulty present on this test, and that it was an appropriately challenging and moderately to easy-to-complete test for the age group assessed. In addition to this, the discrimination values for each item on the test ranged from 0.16-0.69, with an average for all items of about 0.44. The majority of the test items were good discriminators between students who were proficient at algoritmic thinking and those who were not; therefore the items with lower discrimination values were retained due to their importance in establishing the test's content validity.

Second, descriptive statistics confirm the valid use of Tech Check 2. The results of the Turkish adaptation research show that the average score of third-grade students was 10.76 (SD = 2.87), and the scores varied between 1 and 15. These findings confirm that Tech Check 2 is sensitive to individual differences in algorithmic thinking among early primary school students.

Tech Check 2: a tech check specifically suited to the current study for several reasons. Firstly, it is intended for and validated with early primary schoolchildren of the same age as

the target participants of the current study. Secondly, in its unplugged and graphical form, it enables the assessment of skills in algorithmic thinking in a way that is not dependent on previous experience with tech, thereby increasing the internal validity of the study. Lastly, the skills encompassed in the tech check—sequencing, planning, logical thinking, and debugging—mirror the skills targeted in the proposed intervention of the 2D directional logic game in the current study. On balance, features such as the presence of empirically supported reliability coefficients (KR-20/Cronbach's $\alpha \approx 0.70$), item difficulty and discrimination values, defined methods of scoring, and a strong fit with an emphasis on instruction during an intervention program justify Tech Check 2 as a sound instrument for measuring primary outcomes for this investigation.

| Grade Level | N | Mean | SD | KR-20 (≈ α) | Avg. Item Difficulty | Avg. Item Discrimination | Interpretation |
|---|---|---|---|---|---|---|---|
| Grade 3 | 244 | 10.76 | 2.87 | .70 | .72 | .44 | Appropriate difficulty, good discrimination |
| Grade 4 | 238 | 12.28 | 2.38 | .67 | .82 | .34 | Too easy; ceiling tendency |

*Values are summarized from the original item and descriptive statistics reported by Çetin, Şendurur, and Otu (2022) for the Turkish adaptation of Tech Check 2 (see Tables 6–9 in the original study).*

## Data Collection Procedures and Intervention

The collection of data will adhere to a structured process of pretest–intervention–posttest sequences. The implementation of all procedures will occur during school hours in the computer lab or classroom environment using school tablets/computers. Both test groups will utilize the same assessments and participate in equal time for play and instruction to allow for continuity. Tests will be done paper and pencil, control group will take traditional lessons with no computer usage while the experimental group uses tablets or computers to play the assigned game.

Since the pre-test and the post test is TechCheck2, we considered the testing threat. To overcome this internal validity threat, we won’t be announcing their pre-test results and won’t be sharing the correct answers until the end of the post-test.

### *Ethical Considerations*

After the participants are recruited, we'll have their parents' informed consent since we're working with the children. Also child's assesnt, anonimity and confidentiality of data will be considered.

***Pretest Procedures***

As part of the pretest process, all children participating in this study will complete a standardized assessment (Tech-Check 2 (Turkish Version)) designed to measure foundational algorithmic and computational thinking skills during the first session of the study. The instrument will be administered during a single session; individually in a quiet classroom or computer lab environment. Total length for completion will be 35-40 minutes.

***Intervention Overview (3/week × 4 weeks = 12 sessions)***

After the pretest, students will be divided into two groups. The first group, the experimental group will play a targeted Sokoban puzzle game (Thinking Rabbit Inc., 1982). and the second, the control group will receive traditional, non-game-based instruction targeting foundational algorithmic thinking skills.

All students will participate in their assigned condition three times a week for a total of four weeks and each session will last approximately 20 minutes. We'll take their attendance at the end of the sessions. All sessions for both groups will be conducted under comparable classroom conditions to ensure consistency.

***Experimental Game Intervention: Sokoban***

The experimental group will play Sokoban, a two-dimensional puzzle video game where players move a character one step at a time across a grid while pushing objects to a location on the grid, while at the same time avoiding or negotiating around the constraints of the grid itself. Through the gameplay experience, Sokoban encourages children to think sequentially, plan a path with multiple steps ahead of time, anticipate the result of an action or series of actions they have taken, decompose a task into smaller parts, and finally, debug errors.

These skills align with the constructs measured by Tech-Check 2. Even though the Sokoban game is not an educational coding game, its framework and mechanics encourage children to use the same logic and reasoning approach to developing algorithms used for solving algorithmic problems. Players of Sokoban will start with uncomplicated beginning levels and progress through levels quickly as they learn the gameplay mechanics.

- **Week 1 (Levels 1–3):** Children's introduction to game mechanics and the sequential nature of actions will be illustrated through levels that are based on one-dimensional movements (i.e., left/right, up/down) and simple actions taken in sequence.
- **Week 2 (Levels 4–6):** children will build on the skills that they learned in week 1 by incorporating the concept of "path planning" (paths created in 2D space as opposed to 1D) into their learning. This will allow children to take the anticipated outcomes of a move into consideration when making their next move, choosing the path of least resistance.
- **Week 3 (Levels 7–9):** Emphasize the importance of identifying errors in problem-solving through levels that are designed to require tight spacing and walls separating the corridors. This will require that children learn how to revise strategies based on error recognition and therefore break down an overall task into simpler pieces.
- **Week 4 (Levels 10–12):** Serve to consolidate previously learned concepts and skills in addition to continuing to develop problem-solving ability.

This structured progression was designed to scaffold sequencing, planning, constraint-based reasoning, and debugging skills in a developmentally appropriate manner for children aged 7–9.

### *Control Group Intervention: Traditional Instruction*

In order to assist participants in developing foundational algorithmic thinking abilities, the control group will partake in traditional, teacher-led lessons that offer no use of digital games. All lessons will last as long as the sessions in which participants utilise the games. They will concentrate on abilities such as: sequencing; step by step planning, basic problem solving. The educational materials will be carefully organised and appropriate for the participants' developmental ages, and increase in difficulty proportionately.

Potential instructional methods include: paper-and-pencil exercises, unplugged algorithmic tasks (i.e., following & constructing step-by-step instructions) examples demonstrated by the teacher.

### *Fidelity of Implementation*

To ensure all participants receive consistent instructions (i.e., "All participants in Groups 1 and 2 will be given the same general instructions about effort and participation, not performance.") and that teachers will not provide any assistance to their students on how to

play the game strategically but can assist them with any technical difficulties that may arise due to the nature of the game.

All participants will be observed by either the researcher or the classroom teacher and their attendance and other observable behaviours will be recorded for each session. These fidelity checks were conducted to ensure that students were present and actively participating throughout each session. Fidelity data were not used quantitatively unless there were observable deviations that could directly affect the validity of intervention outcomes (i.e., continual inattention, disengagement, or not completing the task as assigned). Fidelity records were used to track attendance in these situations and to indicate sessions or individuals whose data needed more careful review in the analysis.

### *Posttest Procedures*

Upon completion of the four-week intervention, students will complete the same assessment used during the pretest: Tech-Check 2. To ensure no variability due to external factors, the procedures for administering the assessments, instructing students prior to taking the assessments, and setting the testing environment will be the same as they were at pretest time. Approximately 35–40 minutes will be the length of time required to complete the posttests.

### *Data Preparation*

Scores for Tech-Check 2 will be derived by adding up the number of responses correctly identified by the student to the presented questions.Pretest and posttest scores will be prepared for analysis by examining individual student gains across the intervention period. Data will be screened for completeness prior to analysis.

## Data Analysis

The first stage of data preparation is to generate both pre-test/post-test raw scores of each participant. The main dependent measure, "gain scores," will be determined by subtracting pre-test from post-test scores (Gain Score = Post-Test - Pre-Test) for each student, and descriptive statistics such as means, standard deviations, and skewness will be computed for gain scores for experimental and control groups.

For inferential statistic analysis, independent samples t-test analyses will be performed to compare the mean gain scores of students in the experimental group who received 2D directional logic game intervention and students from the control group who

received traditional instruction. Dependent samples t-tests will also be performed within each group to examine the changes in pre and post tests. For t-test analyses, it will be important to consider the following assumptions about the data: a) the data have a normal distribution b) the observations are independent of each other as defined in the study design c) the scores for the assessments are measured on an interval scale. The alpha level will be set at $p \leq 0.05$ to determine whether or not the statistical results are significant.

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Wong, G. K. W., Jian, S., & Cheung, H.-Y. (2024). Engaging children in developing algorithmic thinking and debugging skills in primary schools: A mixed-methods multiple case study. *Education and Information Technologies, 29,* 16205–16254. https://doi.org/10.1007/s10639-024-12448-x

Yang, T.-C. (2024). Enhancing elementary school students' computational thinking and programming learning with graphic organizers. *Computers & Education, 209,* Article 104962. https://doi.org/10.1016/j.compedu.2023.104962

Zengin, E., & Karal, Y. (2024). Developing a game-based test to assess middle school sixth-grade students' algorithmic thinking skills. *International Journal of Assessment Tools in Education, 11*(1), 88–108. https://doi.org/10.21449/ijate.1327082

**PART2**

# Appendix

## Appendix A

## TechCheck2 (Turkish Adaptation)

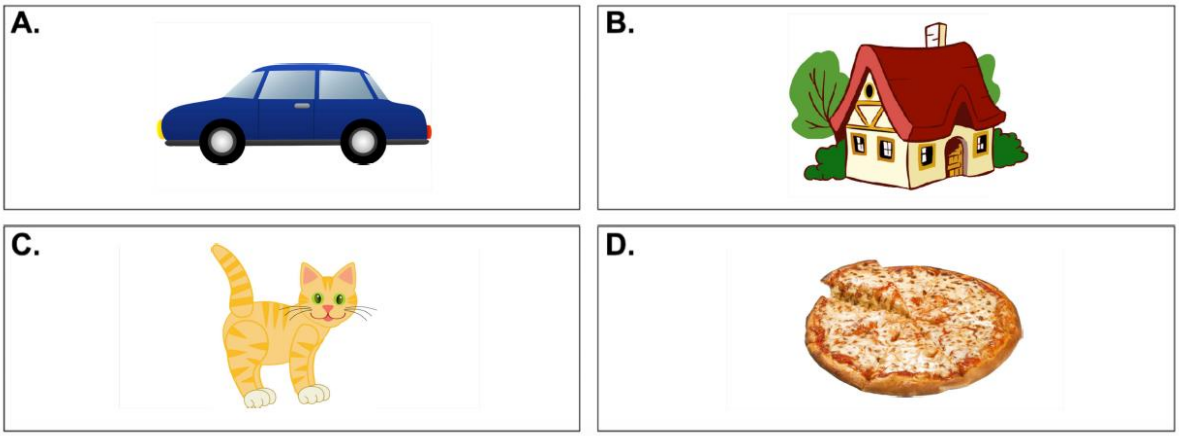
**Instrument Source:** https://drive.google.com/file/d/1R6Yhfr-7uugikO1TCHlKvHEF54DkrzDP/view?usp=drive_link

Appendix A contains examples of items found in the Turkish version of the TechCheck2 assessment. These example items illustrate the types of questions asked in the assessment to measure foundational computational and algorithmic skills. The full assessment was administered according to standardized procedures.

### Sample Item 1. Object Classification

Örnek Soru 1

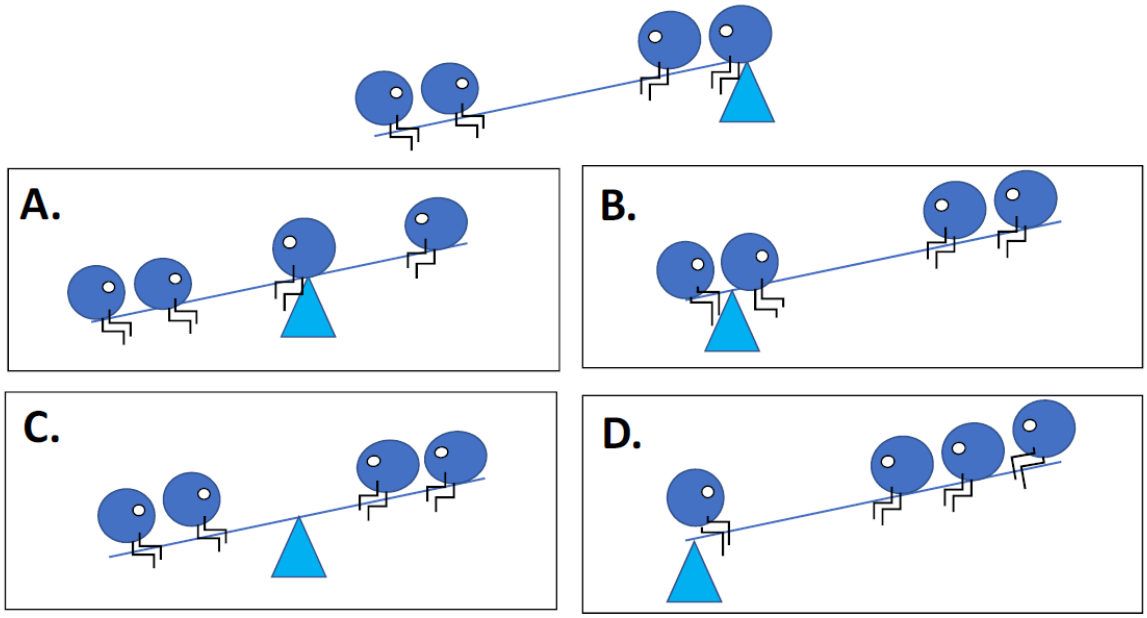
Aşağıdakilerden hangisi bir hayvandır?



**Sample Item 2. Mechanical Reasoning (Balance / Seesaw)**

-4-

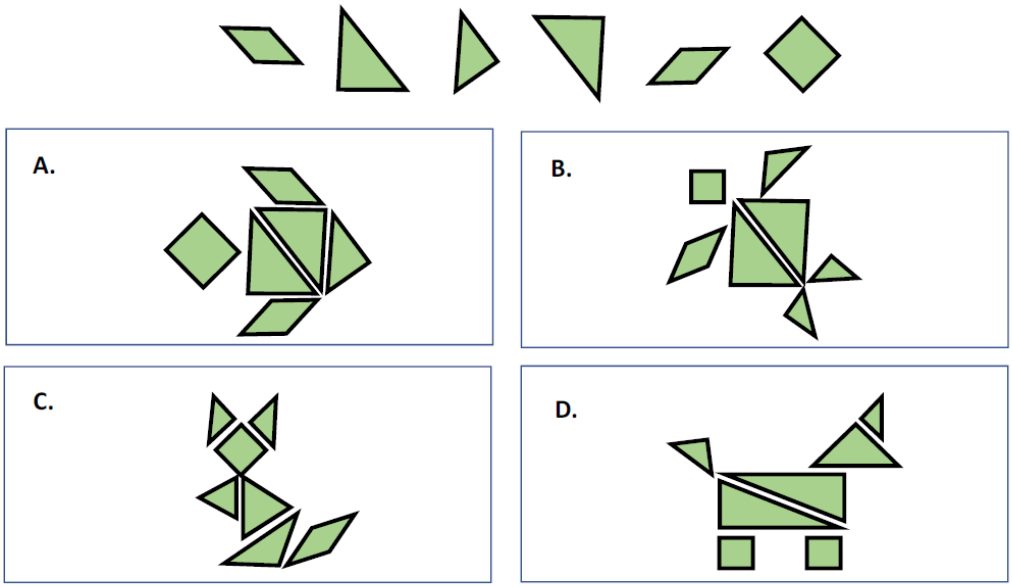
Aşağıdaki tahterevalli aşağı yukarı hareket edemez. Çalışması için nasıl değiştirilmelidir?




**Sample Item 3. Shape Composition**

-5-

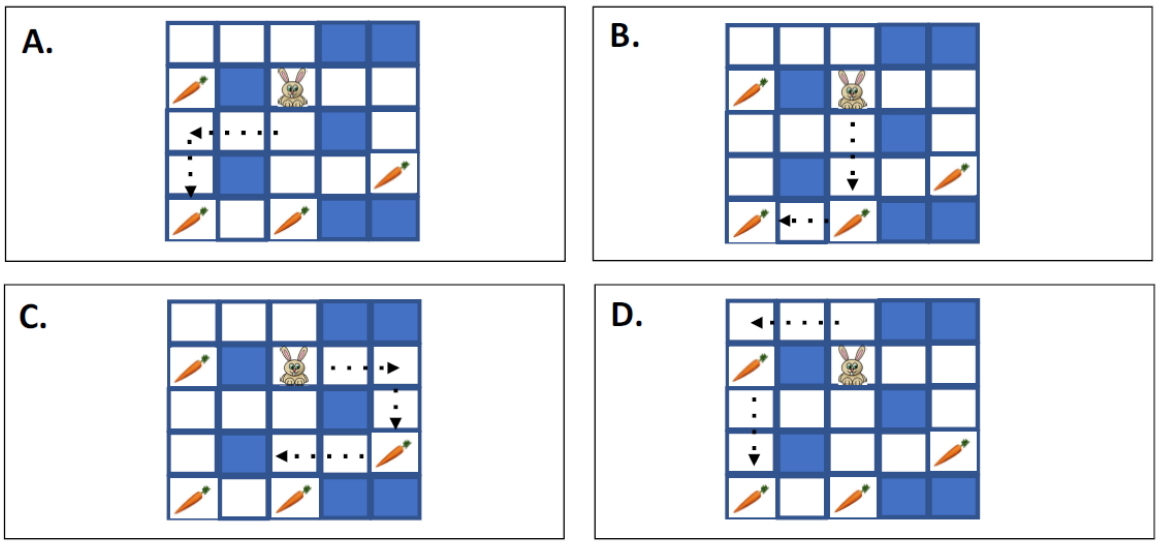
Aşağıdaki şekillerin tamamı yalnızca bir kere kullanılarak hangi hayvan şekli elde edilebilir?



**Sample Item 4. Rabbit Path**

-9-

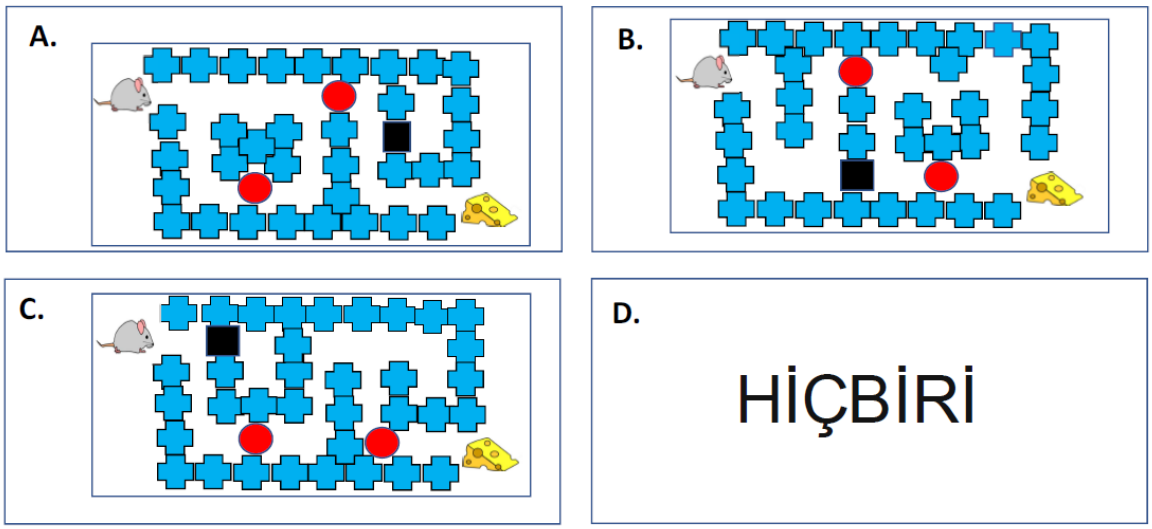
Tavşancık bir seferde sadece bir kare zıplayabilir.
Tavşancığın **iki** havuç alması için en hızlı yol hangisidir?



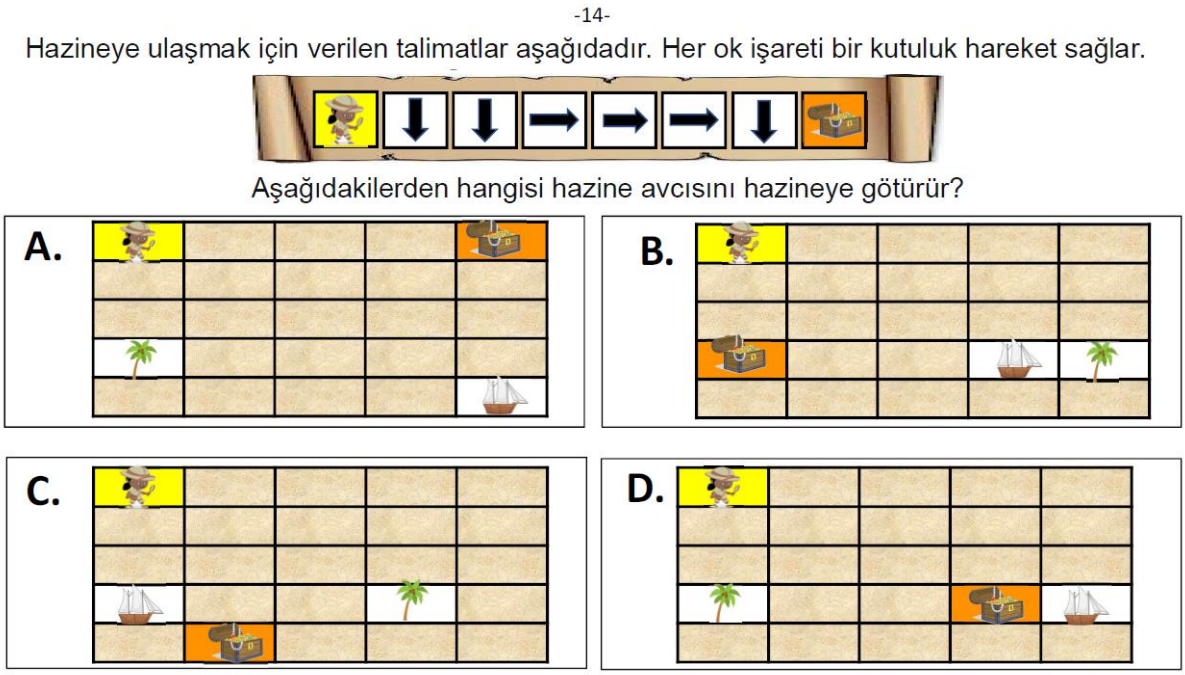

**Sample Item 4.1. Rabbit Path**

-15-

Fareler mavi duvardan veya kırmızı ışıktan geçemez. Fakat siyah tünelden geçebilir.
Aşağıdaki hangi fare peyniri alacaktır?



**Sample Item 5. Maze / Treasure Hunt**




## Appendix B

### Description of the Sokoban Game Used in the Intervention and Level Progression

Sokoban is a puzzle that involves moving boxes through a grid to their designated locations. The player moves by taking small steps in one of four directions. The goal of each level is to maneuver the boxes without hitting obstacles like walls or narrow corridors. In order to do this successfully, players must carefully consider the order of their moves; players cannot pull the boxes and making mistakes can cause them to create an unrecoverable state, requiring the player to reset and rethink their move strategy.

The Cognitive processes involved in Sokoban were found to correlate strongly with the major cognitive processes underlying algorithmic thinking skills. As such, Sokoban was included in our study as an example of game design that represents the process of sequential reasoning, path-planning, and constraint-based problem solving and error correction.

Week 1- Level 1–3: Sequencing, basic directional movement

Week 2- Level 4–6: Path planning, anticipation of outcomes

Week 3- Level 7–9: Debugging, error correction

Week 4- Level 10–12: Strategy consolidation, problem solving

**PART 3**

## Annotated Bibliography

Dlab, M. H., Hoic-Bozic, N., Andelic, M., & Boticki, I. (2020, April). Digital games and tools for development of computational thinking in primary school. In *Proceedings of the International Conference on Management, Economics & Social Science (ICMESS 2019),* Brussels. University of Rijeka & University of Zagreb. https://degames.uniri.hr/wp-content/uploads/2019/08/ICMESS_Brussels.pdf

This paper reviews the potential use of games and digital tools to foster Computational Thinking and students' programming abilities, with a particular attention to primary school age. Authors present an overview of some free examples suitable for this school level and describe a model for the development of CT skills designed within the project GLAT. This study advocates for encouraging CT from primary school, given that such skills are considered fundamental in the 21st century and useful in almost every career. Fundamental skills of CT concern decomposition, pattern recognition, abstraction, and algorithm development, while basic concepts regarding programming concern sequence, loop, variable, and conditionals.

As a conceptual and review paper, no students were directly involved in the study. The target population of the educational implications and resource analysis was lower-grade primary school students. This is because these children are growing up surrounded by digital technology, making digital games a natural and motivating way of learning for them. The paper discusses the need to educate primary school teachers in integrating games into different subjects to develop these skills.

This paper is based on a literature review and conceptual model design. The authors analyzed the potential of GBL strategies-situations where games have predefined learning outcomes-and gamification, or adding game elements into non-game contexts, to support the development of CT. Considering this, the design compares the integration of games as either unplugged activities that involve physical objects or movements, or digital forms, which today are more relevant for current students. This paper also discusses the GLAT model, which applies blended e-learning, PBL, and IBL to the training of teachers in these game-based scenarios.

Data for this paper were mostly secondary and included samples of digital games and tools freely available on the Web. The tools researched in the study involved the following logic puzzles: Sudoku, Jig Saw Doku and Kakuro; block-based programming games, namely Code Combat, LightBot, and Bee (Code.org). These tools were assessed to see if they supported the required computational skills of decomposition, pattern recognition, abstraction, and algorithm design. Reports regarding learning scenarios developed by teachers within the GLAT project were also employed in the research study.

This paper concluded that digital games are a great opportunity for educational purposes. Also, learners could start learning basic programming concepts and develop their computational skills early in primary school. Key findings include:

- **Game-based learning (GBL)** is shown to educate by stimulating creativity, enhancing concentration, and developing abilities for problem solving, logical reasoning, and strategic thinking.
- **Digital relevance** means the game fits into the interests of modern children, corresponds to new technologies surrounding them, and improves attention span and spatial vision.
- The paper focuses on the ability of the teacher to make the games or tasks, like puzzles or mazes, part of an **algorithmic process** which later on can be applied in other contexts.
- In terms of **future outlook**, integration of games may yield better attitudes towards programming and interest in STEM careers among students.

This article is fundamentally essential because it provides the overarching theoretical justification and pedagogical context for our whole quasi-experimental study.

- It introduces digital games and GBL as valid, appropriate, and useful tools in teaching primary school with the ability to develop algortihmic and computational thinking. Games like Bee and Run Marco! entail directional logic as a tool for learning in the area of sequence and loops.
- Paper defines and focuses on algorithm development as one of the four basic CT skills that can be developed in primary school, thus supporting the selection of Algorithmic Thinking as the main dependent variable in our study.
- RQ5 mainly addresses unplugged/transfer: the discussion of integrating both digital/plugged and unplugged activities is very relevant to the scope of your study,

since it gives a framework for discussing the potential transfer of skills learned in our 2D game to AT tasks.

- Also, methodologically, it provides examples of ready-made game tasks-like the Bee game task presented in Figure 1-similar to the directional-logic problems our proposed game will incorporate, supporting the development of our intervention materials.

Falloon, G. W. (2024). Advancing young students' computational thinking: An investigation of structured curriculum in early years primary schooling. *Computers & Education, 216*, Article 105045. https://doi.org/10.1016/j.compedu.2024.105045

Falloon (2024) explored how a planned problem-based class for CT, with the support of the teacher, can enhance the capability for CT in young primary students. The study involved seven lessons where the students, with the use of small floor robots, carried out basic CT tasks, including the making of step-by-step commands, mistake fixing, finding patterns, and thinking simply to solve problems. Initially, the teacher modeled for the students what to do, explained examples, and helped them understand how the robots move. Later, students tried solving problems more independently and worked in small groups discussing different ways of making the robot go to the right place. This research utilized an intervention-based design to monitor the progression of student's development over time (throughout the lessons), rather than simply before and after the intervention. Data collection consisted of observational data from classroom activities, as well as artifacts produced by students when utilizing robots and student conceptual change within the context of the task (to support if a class activity was unsuccessful). Among other findings, the researchers found that through the use of robots, students improved their ability to write clearly and ordered instructional steps while also recognizing patterns they could utilize as guides for completing tasks more quickly, and they could easily modify incorrect behaviors of robots as needed to achieve the desired outcome. Several students also expressed higher levels of thinking by applying previously learned strategies to unfamiliar tasks, articulating their process of thinking through pattern recognition, developing efficient and/or optimal robot paths through pattern recognition. According to the findings of this study, children as young as 6 years of age possess the ability to create computational thinking and programming skills provided they receive sufficient support and time to do so. This assumption runs contrary to the

prevailing belief that children of such a young Division of Early Childhood Education (DEC) have insufficient ability to understand the principles of computing in their early development stage.

Additionally, this research indicates that implementing an intentional CT instructional model with effective teacher involvement produces better student learning outcomes than allowing students to simply explore or engage freely without an instructional plan. The instructional model fosters student's ability to construct concepts sequentially, thereby ultimately understanding the rationale behind the functioning of the CT model. The study also showed that in a short period of time, CT skills can grow once students have practiced again and again and get feedback from the teacher. Overall, this research says that providing young children with structured CT lessons helps them develop more confidence to solve simple and bigger problems in an organized way, and prepares them for future learning in computer science, or other school subjects, where logical thinking plays a significant role.

It is relevant to my topic since this article also investigates how young children can improve basic algorithm and thinking skills through activities in the digital environment. My work looks at how a 2D directional-logic game helps children aged 7–9, and this article works with young students too. It was found that digital tasks can be helpful for learning things such as sequencing and problem-solving by children, which are significant parts of algorithmic thinking. Although the article does not involve a 2D game, it nonetheless supports the idea that game-style activities can help students think better. Thus, this article helps me to understand why using simple digital games can be useful for teaching algorithm thinking to young children.

Giannakoulas, A., & Xinogalos, S. (2023). Studying the effects of educational games on cultivating computational thinking skills to primary school students: A systematic literature review. *Journal of Computers in Education*. Advance online publication. https://doi.org/10.1007/s40692-023-00300-z

This study aimed to investigate and describe old research on how educational games help primary school children learn basic computational thinking skills. The authors themselves did not work with students but investigated 61 studies that were already conducted with primary school students who used game activities for learning. A systematic review of the literature was conducted, entailing a search through large academic databases

and selecting the papers using special rules. The chosen ones were then organized by the CT skills taught, the ideas of programming used, the kind of games included, the duration of the lessons, and where this learning took place. To gather data, information was taken from each study: age of children, goals of the lesson, gaming tools used, and results of learning. Further, they noted the tools used to measure learning, such as small tasks, tests, teacher notes, and questionnaires. Different studies used different methods of measuring CT, sometimes making comparisons hard.

Generally, most findings provided evidence that educational games supported the improvement of CT skills in students, such as sequencing, problem-solving, abstraction, algorithm thinking, and simple coding ideas. Quite a few children became happier and more motivated to learn with games, and thus they enjoyed programming more. Sometimes games made students work together and try again after mistakes, which helped learning. Some problems arose: most studies had small groups, short lesson time, and weak research design, so results were not always strong. Most of the studies, according to the authors, looked only at basic CT skills, not harder ones. They also proposed new research should use better methods, longer time, stronger CT tests, and look at more CT skills in order to understand better how to help young children learn computational thinking.

This article is related to my topic because it explores how educational games help primary school children improve in computational thinking skills. My study also wants to see whether a 2D directional-logic game can help children aged 7-9 learn basic algorithmic thinking. The article mentions that most game-based studies help students improve in skills like sequencing, problem-solving, and simple algorithm ideas, which are similar to the skills that I want to research. It also shows that games make learning more fun and help children stay motivated. That will support my idea that using a game could be a good method for teaching algorithmic thinking. In the article, there was some mention of what was done before and what is missing; this helped to explain why my study would still be important. Overall, the study gives background on how educational games can help young students learn thinking skills, so it is clearly connected to my topic.

Gunderson, S. W., & Lampropoulos, G. (2025). Using serious games and digital games to improve students' computational thinking and programming skills in K-12 education: A systematic literature review. *Technologies, 13*(3), 113. https://doi.org/10.3390/technologies13030113

The focus of this study was to perform a systematic literature review examining the impact of digital games (serious games and otherwise) on K-12 students' computational thinking and programming skills. The goal was to compile information from previously published research between 2011 and 2024 concerning the characteristics of these studies, the advantages and difficulties of game based learning, and the essential principles for developing optimal educational games in this particular domain. The authors of this article indicated that the approach used was a formal Systematic Literature Review following the PRISMA framework for thorough and rigorous identification and screening of data. Articles were found in the IEEE, Scopus, and Web of Science databases after extensive searching via a structured query, leading to the final selection of 78 articles which comprised the participant data pool for this literature review. A classification of these articles based on methodology divided the content into three categories: "Theoretical and Review Studies" (9 articles), "Proposal and Showcase Studies" (11 articles), and "Experimental and Case Studies" (58 articles). The authors noted that 89.7% of the experimental studies classified the student participant group as their primary source of data collection.The authors of the article examined all 78 articles reviewed and analyzed how data was collected in the study. Most studies (82.8%) were conducted using quantitative and combined methodologies and the most often used data collection instruments were surveys and questionnaires. Survey and Questionnaire instruments were used in 82.8% of studies classified as experimental, along with tests, interviews, and observational instruments. The systematic review indicated that ad hoc tools were used frequently, but there were also standardized tools, including the "Computational Thinking Test."

In this regard, the major conclusions of this article are consistent with prior research that shows the positive educational impact of game-based learning practices. The systematic literature review found that both serious games and digital games are meaningful educational tools. Serious games and digital games effectively support and improve computational thinking and programming skills, and serious and digital games are viewed positively in educational settings. The two most frequently cited benefits identified by researchers were:

50% of authors who conducted experimental studies reported that serious and digital games increased student learning performance; 50% reported that students enjoyed learning more by using serious and digital games; and 50% indicated that they were more motivated to learn. Enhanced levels of student engagement, enhanced levels of student confidence, and immediate feedback about the student's progress are also listed as positive aspects of using games as instructional tools. Because this article provides a systematic literature review on game-based learning, it provides a wealth of evidence that supports the reasons for using game-based learning to inform our research study.Additionally, we support our 2D game focus, because it has conducted experimental studies using 58 of the 2D Game. They are identified as the most frequently used type." Also, it is consistent with out target population and identifies successful game mechanisms.

Hsu, C.-C. (2018). Applying game mechanics and student-generated questions to promote algorithmic thinking skills. *Computers & Education*, *144*, Article 103698. https://doi.org/10.1016/j.compedu.2018.02.002

The main purpose of the present study was to test whether using game ideas in concert with questions prepared by students can help students improve in algorithmic thinking. The subjects were junior high school students in Taiwan participating in a computer-science class. Most of them did not know much about algorithm problems before, so they were beginners. The research design was a quasi-experiment. Students were divided into two groups: an experimental and a control group. The experimental group learned with game elements like points, levels, badges, small time challenges, and fast feedback. They also made their own questions to review new lessons. The control group only learned by normal teacher lecture and did not do the game or question-making activities. The data was gathered using algorithmic-thinking tests, examination of the student-made questions, observing the class to see how the students behaved, and questionnaires to check motivation and learning attitude.

In this respect, the experimental class performed better in algorithm thinking, such as identifying patterns, making steps, putting actions in order, and understanding simple logic. That means game mechanics and question-making contributed to students' thinking more and learning better. The experimental students talked more in class, tried to solve problems more, and showed more interest and motivation. When making questions by themselves, students

reviewed the course contents again, reflected on mistakes, and understood ideas more deeply. Sometimes, they shared and discussed questions with classmates, which contributes to learning from classmates and being more confident. The class observation showed that during class the students of the experimental group cooperate more and try more different ways of solving a problem; because of this, they practice more and understand algorithm thinking from different sides.

The results indicated that student-made questions have helped students find out what they know and what they do not know. Game elements, such as levels, badges, and feedback, make learning more fun and not boring. Many students have said that this game style of learning is more interesting, making them feel more active to help them stay focused during class. In conclusion, the research indicated that the combination of game mechanics used in conjunction with question creation produced an environment that would support the development of algorithmic thinking and provide for a higher level of student self-confidence, cooperative learning, and a greater interest in computer science.

This article is relevant to my topic because it also investigates how game-based learning can enhance students' algorithmic thinking. My research looks into the playing of a 2D directional-logic game by 7-9-year-old children, and this article shows that using the ideas of games can make students think better about steps, patterns, and simple logic. Even though students in this article are older, still the very idea is similar: games and special activities help learners understand problem-solving. The article also points out that game elements raise motivation and make students more active, which for my topic is important because my game also aims at maintaining young children's interest. Also, it has been shown that students learn much better when they create questions and think, which means that the idea of interactive game activities can enhance algorithm thinking. Because of that, the article supports with useful findings and background my study related to the improvement of basic algorithmic thinking in young primary students by means of a 2D logic game.

Kanaki, K., & Kalogiannakis, M. (2022a). Assessing algorithmic thinking skills in relation to age in early childhood STEM education. *Education Sciences, 12*(6), 380. https://doi.org/10.3390/educsci12060380

In this quantitative study, the researchers investigated whether or not there was a correlation between algorithmic thinking skills and the students' age during their first two

years of primary school. More specifically, they were trying to find out whether or not there was evidence for algorithmic thinking being developed in young children of such a limited age (i.e. 1st and 2nd graders). Earlier studies had generally investigated algorithmic thinking in much broader age ranges and did not consider the smaller ages that the researchers were studying. The research was conducted within the framework of a STEM curriculum, specifically an Environmental Studies class, with an emphasis on the incorporation and connection of computational thinking with other fields within science. The sample consisted of 435 students who were from Heraklion and were enrolled in the school year 2018-2019. It was a balanced sample with regard to both grade level and gender. The authors point out that in Greece, primary education begins at six years of age, making the age range of this sample fall under early childhood development, i.e., ages 6-8. The authors used a cluster sampling technique to get a representative sample of the socio-economic status of the students. This study was designed to be quantitative in nature and focused on testing the hypothesis that "Algorithmic thinking skills are NOT associated to students' ages in the early primary school years". Each intervention lasted 30 minutes for each individual class, and multiple statistical techniques (chi-square tests, odds ratio calculations, and ordinal logistic regressions) were performed on the results to investigate any correlations. The data was gathered through an innovative platform called PhysGramming, which allowed students to create jigsaw puzzles as their form of assessment. The results of the study show a statistically significant correlation between age and the ability to program algorithms, indicating that age is a significant predictor of algorithmic thinking in this small 2-year age range. The results of the study also demonstrate that second graders are more likely than first graders to have satisfactory and excellent algorithmic thinking skills whereas first graders have a much higher chance of having a medium algorithmic thinking ability level. I believe this article will be beneficial to our study due to its direct association with our population and measure, as well as identifying age as a very important control variable for predicting algorithmic thinking.

Kanaki, K., & Kalogiannakis, M. (2022b). Assessing algorithmic thinking skills in relation to gender in early childhood. *Educational Process: International Journal, 11*(2), 44–59. https://doi.org/10.22521/edupij.2022.112.3

The goal of this research was to identify if there is a correlation between a student's ability in algorithmic thinking and their gender, during their early years of education in Greece. The impetus for doing this research was the inadequate representation of women in the Science, Technology, Engineering and Mathematics (STEM) fields, so the authors were hoping to see if the gender gap was already emerging during the first two years of the compulsory education system. The second aim of this research, although still important, was to develop and implement a new tool to assess the development of children, as they feel that there is a significant gap in the assessment of algorithmic/computational capabilities of very young children who are still acquiring reading and writing skills. The study was carried out in Heraklion, Greece, with a sample size of 435 first- and second-grade students, which was balanced between the two grades: 218 were in first grade and 217 were in second grade. Of the total students included in the sample, 210 were female and 225 were male. The sample was selected through the use of a cluster sampling method to include schools with varied socioeconomic backgrounds, making the overall sample more representative of the population. The study has been designed as a quantitative research study. The authors created their data collection instruments, which were based on a digital platform (titled "PhysGramming") to be used in conjunction with an environmental study course. This tool allows students to complete and create digital puzzles at various levels of difficulty through a play-based method. The platform developed log files that automatically recorded the maximum difficulty level accomplished by each student to measure their algorithmic thinking skill.

In summary, the results of this study rejected the hypothesis that there is an effect of gender on algorithmic thinking within this age group. The chi-square analysis provided no evidence of a statistically significant relationship between algorithmic thinking and gender. The authors determined that while there may be gender gaps in STEM fields as students progress through later years of schooling, those gaps are likely not due to intrinsic differences in fundamental ability but rather to societal influences, such as social stereotypes and norms, which have a larger impact in earlier grades than in later grades. This article applies directly to our research because it presents a case study of a two-dimensional (2D) digital game.

Specifically, the 2D puzzle game component of the platform provides a clear, specific case of a 2D game being used in an educational context to evaluate algorithmic thinking skills.

Kovtaniuk, M. S., Shokaliuk, S. V., & Stepanyuk, A. N. (2025). Game simulators as educational tools for developing algorithmic thinking skills in computer science education. *CTE Workshop Proceedings*, 12, 19–62. https://doi.org/10.55056/cte.926

This article examines the growing trend of using game simulations as a teaching method. Through an extensive literature review, the researchers aim to compile and organize the current research on how various game simulators can help to develop algorithmic thinking in students who are studying computer science. They hope to provide a solid theoretical framework for the use of game simulators in education by linking design principles of gamification to cognitive learning theory to explain why these tools are effective. Additionally, they wish to offer real-world examples of how to effectively implement game simulators and the scaffolding and assessment methods that are appropriate for each grade level. Because this is a review of existing literature rather than a study that recruited participants, the authors' data set consists of an extensive body of literature related to the use of game simulators as pedagogical tools, covering the entire spectrum of learners, from primary to postsecondary. As a review of literature and an evaluation of the current research in the field, the authors' methodology incorporates a comprehensive analysis of all available literature on game simulators to identify the trends, benefits, challenges, and gaps in research that exist. The researchers' comparative analysis of various types of game simulators serves as a case study that helps to support their theoretical analysis of the topic. Authors compared various computer programming platforms such as block-based platforms like Blockly Games and Scratch, narrative-driven games like Rabbids Coding and 7 Billion Humans, and open-ended programming environments like Minecraft Education Edition and Kodu Game Lab. They found that if implemented properly, game simulators can significantly increase student engagement, motivation, and development of algorithmic thinking. The authors also state that different types of game simulator tools are most effective for achieving different purposes. For example, block-based programming environments are well suited to teaching basic programming concepts to students; narrative-based games put programming challenges into context; and open-ended programming environments enable creative ways to solve programming problems. The authors also highlight the importance of how students are served through the use of digital tools, stating that students must be given support via

pedagogical scaffolding, teacher support, and multimodal assessment for digital tools to be effective. The identified areas for further research as well as gaps are also noted, with an emphasis on the need for longitudinal studies exploring skill retention (the continued use of a skill after its initial acquisition) and the difficulty in establishing the transfer of learned skills from game contexts to non-game contexts. I believe that because this is both a theoretical analysis and a review of the body of literature related to the field, it serves as a substantial theoretical base for our research study. Additionally, the review analyzes types of games that are specifically applicable to our research and provides age appropriate implementation methods aligned with our targeted age range.

Piedade, J., & Dorotea, N. (2023). Effects of Scratch-based activities on 4th-grade students' computational thinking skills. *Informatics in Education*. https://doi.org/10.15388/infedu.2023.19

This quantitative study examined the impact of Scratch-based activities on the development of CT performance along with its related concepts during one school year among Grade 4 students. The authors tested Hypothesis $H_1$ that participation in Scratch-based activities positively and statistically significantly enhances students' CT performance. Besides, the researchers explored how demographic attributes like gender and age ($RQ_3$), and students' academic achievement in Mathematics, Science, and Portuguese influence students' CT performance. Interest in integrating CT into school curricula and empirical evidence regarding how to effectively deploy this type of intervention in elementary education have motivated this research.

This study initially enrolled a sample of 189 participants, all 4th-grade students, from two clusters of public schools in the Lisbon district of Portugal. Children were usually 9 or 10 years old. Students were assigned to either an Experimental Group (N = 96) or a Control Group (N = 93) because of institutional constraints. According to the children's teachers, none of the students of either group had previously worked with Scratch or similar activities of programming and development. The comparative socioeconomic, demographic, and academic characteristics of the two school clusters allowed for a well-matched comparison between groups.

A quasi-experimental design has been used that comprises quantitative measures. Randomization of students within the educational setting was not possible; therefore, a non-

probability sampling method has been used. It is a post-test-only design. Since none of the groups had prior programming experience, such a design was appropriate. Students in the Experimental Group received weekly programming instruction (through Scratch) for one hour each week throughout the 2020-2021 academic year, whereas those in the Control Group did not participate in programming lessons or any related activities.

The principal source of information was the Beginners Computational Thinking Test (BCTt), a validated and direct psychometric assessment tool for students aged 5 -12. There are 25 multiple-choice questions in the BCTt, which are divided into six groups (i.e., sections) that evaluate several areas of CT (Computational Thinking: sequential programming, simple and nested loops, conditionals, etc.), depicted using a maze/canvas style graphic. The final academic results for Mathematics, Science & Portuguese were obtained from each student's main teacher in the experimental group; these results consisted of grades that ranged from 1 - 5. The BCTt presented an internal consistency of $\alpha = 0.85$.

The major finding of this work is that Scratch-based activities are very effective. Effectiveness can be demonstrated in some ways including:

- Comparing the two groups, there was a significant difference in test scores. The above statistics further demonstrate that the increase in scores was due to the programming instruction. Therefore, the group using Scratch scored notably higher in the CT test as compared to the control group. This confirms the main hypothesis that the intervention works. The effect size was considered "high" since $d = 1.16$. Students in the experimental group scored, on average, 4.66 points higher than the control group, which may serve as a point for the boost in their CT skills.
- In both groups, students found questions about sequences and simple loops easier than more complex logic. However, the Scratch group performed much better. The most complex questions concerned conditionals. In these questions, the experiment group outperformed the control group by up to 35%. This evidences that the intervention helped most with higher levels of algorithmic complexity.
- The research did not find any significant influence of gender and age factors on students' performance in the CT test. Additionally, students' scores on their CT tests were not correlated with their final grades in Math or Science; therefore, these findings indicate that the development of this skill was due to programming-based interventions rather than general academic skills.

This paper represents an important methodological reference for our study because it validates almost every aspect of our design. For example:

- **By design**, this is a quasi-experimental design with control group; it has pre/post-test, non-random samples, and uses non-random stratified sampling, just like our proposal idea.
- Scratch is a block-based programming environment. It is all about sequences, movements, and logic. In fact, the logic behind our proposed "2D Directional Logic Game" will be quite similar. The positive results derived in this case surely reinforce the effectiveness of such interactive logic-based tools in fostering algorithmic and computational thinking.
- The BCTt is a maze-style test assesing  sequences, loops, and conditionals. The utilization of the BCTt and the results represent validation for our proposed "directional logic game" that we will use to assess the same fundamental algorithmic thinking constructs.
- **Control Variables:** The finding that neither gender nor age significantly influenced CT performance serves as useful context for our control variables that were age, gender, and prior experience.

Wong, G. K. W., Jian, S., & Cheung, H.-Y. (2024). Engaging children in developing algorithmic thinking and debugging skills in primary schools: A mixed-methods multiple case study. *Education and Information Technologies*, *29*, 16205–16254. https://doi.org/10.1007/s10639-024-12448-x

This study investigates how the computational thinking (CT) skills of children taking a block-based programming course grow over time, specifically algorithmic thinking and debugging skills. The researchers employed a mixed-methods approach to conduct the study. They examined the how this kind of curriculum supports the development of these two CT domains and what is the role of engagement in this process. Engagement was examined in three ways: behavioral, emotional and cognitive. This research offers a significant contribution by demonstrate how these skills develop over time in 'real world' primary schools and not simply as a temporary intervention.

The participants in the study included 191 primary school students from two schools in Hong Kong. At the start of the study, the students were in 4th Grade (approximately 9- 10 years old) and was multi-phased over a couple of years.

For the qualitative component of the study, the researchers also interviewed two teachers and six students (two students from each performance level: high, medium and low) from each school.

This was a mixed-methods, multiple-case study. The researchers utilized both numbers and observations to investigate how students' IT skills evolved. For the quantitative component, algorithmic thinking and debugging were assessed with pre- and post-tests. During the qualitative segment of the study, classroom observations and semi-structured interviews were also utilized to capture students' experiences and the types of learning processes that occurred, and how students participated in this process.

The two full years of research happened during the time programming was included in their regular school curriculum. This research design allowed the researchers to see the quantitative improvements while they engaged in the behavioral and emotional aspects underlying the students' improvements.

Standardized pre- and post-tests were completed by students, with each test lasting about 30 minutes. These tests measured a student's ability to think algorithmically and debug problems.

As part of their research on Scratch programming, the researchers documented various aspects of the participants' behaviours, such as their motivations and how they coded, as well as the reasons for coding in various instructional contexts through the use of observational notes and interviews. Also, after applying their intervention, they conducted semi-structured interviews with both teacher participants and student participants to determine what types of interventions were helpful, what was difficult in terms of engagement, and how engaged students were, as a result of the interventions implemented.

Thus, the researchers' observational notes and interview responses were combined in a meaningful way with their test scores to create greater understanding of students' progress, performance, and potential in programming.

Findings showed a noticeable change in students' IT skills after the two-year program. The greatest improvement was in algorithmic thinking - with an effect size of Cohen's d = 1.077, a large effect size. Their debugging skills also improved noticeably.

However, the shocking find was that girls improved more than boys in algorithmic thinking - giving true perspective, "Could gender influence how IT skills develop?". The

article provided evidence that prior programming experience did not show significant differences in the relationship between prior code experience and subsequent success in learning how to code. All students improved whether they had previously coded or not; thus, support for the lack of an impact of prior coding on their coding course.

Students showed evidence of high behavioral engagement (engaging in learning tasks), emotional engagement (engaged and experienced motivation), and cognitive engagement (applying strategies and problem-solving). The researchers believed engagement was a significant factor in supporting IT skill development, and that structured, active programming lessons can help develop algorithmic thinking in elementary school.

Based on the findings of this study, it appears that the use of structured digital activities is an effective means of providing young learners with opportunities for improving their algorithmic thinking skills.

- Conceptually, both have suggested that algorithmic thinking can be developed in the early years of elementary. Methodologically, their mixed-method, pre/post test research design, and the quasi-experimental design to the research is very similar to ours and makes for a strong reference to our study.
- Additionally, the finding that prior programming experience did not make a significant difference in development was an important finding for us. Because we anticipated that this experience might influence the results when we reflected on our project.
- The participants, ages 9-10, were slightly older than the participants in our study, ages 7-9. However they are in a similar developmental stages. So, they are still a strong reference to us.
- Lastly, we have some key differences as we are have limited time as theirs is over 2 years of duration. Although in terms of participation, the researchers' attention to behavioral, emotional, and cognitive participation provides helpful information for our project.

Yang, T.-C. (2024). Enhancing elementary school students' computational thinking and programming learning with graphic organizers. *Computers & Education, 209,* Article 104962. https://doi.org/10.1016/j.compedu.2023.104962

The goal of this research was to explore whether GO lessons can enhance CT abilities for elementary school students. Besides determining the effect of game formats on students' problem-solving ability, planning steps to solve problems, and finding patterns in the task, this research also explored other ways the GO activities facilitate students' interest and motivations and build their self-efficacy about the class. The respondents were elementary school students who attended school in Taiwan. They did not have substantial prior knowledge of computational thinking and were considered novices learning basic problem-solving abilities for the first time. The method used was a quasi-experimental research design. The samples were divided into two groups, an experimental group and a control group. Experimental group students received GO activities enriched with game ideas, small challenges, points for improving, levels, badges, peer collaboration, and student-generated questions about what was learned. Student self-generated questions would help them to be more thoughtful and double-check their understanding. Students in the control group received regular lessons without games, challenges, or question generation tasks. Data collection involved pre- and post-tests to examine how much the students learn and improve. Classroom observations were carried out to observe students' involvement, cooperation, and behavior during activities. Surveys were also conducted regarding the students' subjective opinions on their own feelings, interests, motivations, and enjoyment about the class. The basic findings revealed that the experimental group outperformed the controls in enhancing CT. They showed more achievement in problem-solving skills, ordering steps to solve problems, identifying patterns, and understanding simple logic. Also, students in the experimental group were active, collaborative with peers, had greater efforts on tasks, enjoyed the lesson more, and endorsed the benefits of question generation in developing thoughtful thinking on the problems, remembering the important ideas, and detecting mistakes they made. This study confirmed the benefits of using games and asking students to generate questions to develop CT in students. It also makes lessons enjoyable and sustains the students' involvement in understanding the concepts better.

This article is related to my topic because it shows that games help children become better at thinking in steps and solving problems. My research is about the playing of a 2D

logic game by 7–9-year-old children to practice algorithmic thinking. According to the article, game activities help students plan steps, find patterns, and solve problems more easily. Even though the age group of students involved in the study is bigger, the notion is similar: through games, learning will be more fun, and the students will be kept engaged in their studies. This research proves, too, that making one's own questions helps students make them think more and deepen the idea. It is like my game, where children need to think and make choices during the process of playing. Thus, this article should help me because the ideas in it support playing games while teaching young children algorithmic thinking.

Zengin, E., & Karal, Y. (2024). Developing a game-based test to assess middle school sixth-grade students' algorithmic thinking skills. *International Journal of Assessment Tools in Education, 11*(1), 88–108. https://doi.org/10.21449/ijate.1327082

In this study, the goal was to create and validate a game focused assessment of the AT skills of middle school students. Very few standardized assessments have been created, because the vast majority of traditional assessments are non-interactive, paper based instruments. The main goal of the assessment was to not only be a digital, interactive assessment that asses AT skills of students but also provide an assessment that based learning outcomes related to programming skills.

This study took place across a few distinct phases and for each of the phases different participants were included. First, we had a small pilot group of 24 sixth grade students that piloted the assessment to provide clarification feedback on the items in the assessment. Then we recruited a panel of eight experts in computer education, measurement, and evaluation to review the content to insure we represented the AT construct appropriately. Lastly, in the larger validation phase we had a group of 252 sixth grade students from our target group take the final assessment we developed. Using a large diverse group, we were see the extent the instrument performed well across various ability levels, and different classroom contexts.

The study utilized a psychometric approach based on quantitative principles - focused on the development and validation of a measure. Development involved a linear sequence of increasingly expansive steps informed by initial theoretical definition of the construct of algorithmic thinking, development of items, subject matter expert review, and pilot testing with a more considerable sample, then field testing with a larger sample. To test construct validity, the research team used what is termed "extreme groups analysis," and compared the

upper and lower extreme 27% of scoring participants by use of independent t-tests. Reliability was determined by reference to Kuder-Richardson Formula 20 (KR-20) measures of internal consistency for binary measures. The research team used Downing's (2006) twelve-stage approach, which included construct definition, item development, pilot testing, and validation with a larger sample.

The main source of data was a new game-based algorithmic thinking (AT) test, which was comprised of 24 test items: 22 real and 2 sample test items. The testing instrument was entirely in a digital format. Students engaged in interactive tasks that included sequencing command(s), creating algorithms, implementing looping reasoning, and applying conditional reasoning. The research team collected students' responses and item-level data for reliability and validity analysis of the assessment instrument.

The results demonstrated that the assessment had both validity and reliability, evidenced by a KR-20 reliability coefficient of 0.83, which indicates good internal consistency. The t-tests, which contrasted the groups scoring at the highest and lowest percentiles for 2 of the 3 game-based measure items, resulted in significant differences ($p < .01$) indicating that the instrument was appropriate to differentiate levels of algorithmic thinking skill. Expert judges met to classify the content of the development test indicated that it appropriately reflected the construct, and matched the students' developmental age. Overall, the study demonstrated an effective means of assessment based on games for measuring algorithmic thinking, and concluded that their measures were also valid and likely more engaging for students than traditional test measures.

This article is essential for providing the psychometric foundation for using a *game-based test* to asses *Algorithmic Thinking (AT)*, which is central to our proposed instrument section.

- Both studies are focused on algorithmic thinking skills. The core focus on this paper was creating an validating a assesment which directly supports our plan to use a puzzle-based, game-like measure. At the very least, it demonstrates how this can be measured in a fun, game format.
- The test was carefully developed and contained three stages (expert review, pilot testing and reliability testing (ex. KR-20, group comparisons) which provides a substantial example for us to draw on on behalf of our project's pre and post test.
- This study gains more traction in supporting our conception of test reliability in game environments. While participants (sixth grade, 11-12 years old) were older than

participants in our study (7-9 years old) they still offer support for our hypothesis that algorithmic thinking is a reliable measure behaviorally with interactive, digital tasks. Together, this article provides a robust conceptual and methodological base, and further supports, an evidentiary foundation for our project part of using the game-based measures of algorithmic thinking, as we have planned.